pfi

Python for Fixed Income

COURSE 9 · DEPLOYMENT

Model Files and Drift

WEEK 3 · DAY 3

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Today: model files and drift

Today

1. Course 4 day 11's task, served
2. A model file and its card
3. The model behind an endpoint
4. The held-out days arrive as requests
5. Drift: the inputs against the window
6. Flags, and the error by quarter
7. What drift does not say

Two windows, split by date

Development years

2015-01-02 to 2022-12-30, 2,001 days

fits the model

Held-out years

2023-01-03 to 2026-10-02, 939 days

arrive only as requests

A model stops changing the day it is saved. Its inputs do not.

What the service returns

The fitted same-day 30 Yr yield: for the same date as its two inputs, with its window.

Allowed names

the fitted same-day 30 Yr yield

inputs outside the training window

the drift table

Never called

a forecast of rates

a signal or an outlook

a prediction

Every response carries the training window: 2015-01-02 to 2022-12-30.

Course 4 day 11's task, served

Inputs and output

The same day's 2 Yr and 10 Yr par yields in; the fitted same-day 30 Yr yield out.

```
DecisionTreeRegressor(max_depth=4, random_state=SEED).fit(development[FEATURES], development[TARGET])
```

16

leaves: at most this many different values, whatever the input

9.2 bp

mean absolute error on the development years only

2,001

days fitted on, 2015 to 2022

Course 4: a tree cannot go beyond what it has seen. Today it happens in a service.

A model file and its card

```
"name": "the fitted same-day 30 Yr yield"
"features": ["2 Yr", "10 Yr"]   "target": "30 Yr"
"training_window": "2015-01-02 to 2022-12-30"
"rows": 2001
"sha256": "9a6d8fdd1fafbc3468865cec..."
"versions": python, scikit-learn, joblib, numpy
```

model.joblib

3,633 bytes

a pickle: loading it runs code

never committed

```
refused, ValueError: model.joblib is not the file its card describes: its SHA-256 differs; do not load it
```

load_model checks the SHA-256 against the card before joblib.load runs.

Never load a model file someone else made

joblib's own documentation

"joblib.load relies on the pickle module and can therefore execute arbitrary Python code."

joblib.readthedocs.io/en/stable/generated/joblib.load.html, read 2026-10-12

Python's own documentation

"The pickle module is not secure. Only unpickle data you trust."

docs.python.org/3/library/pickle.html, read 2026-10-12

The SHA-256 proves a file is yours, unchanged. It never makes a stranger's file safe.

The model behind an endpoint

```
200 {
 "name": "the fitted same-day 30 Yr yield",
 "as_of": "2022-12-30",
 "fitted_30y_pct": 3.9053846153846163,
 "training_window": "2015-01-02 to 2022-12-30",
 "inputs_outside_training_range": []
}
```

POST /fitted-30y

as_of: the date the yields are from

y2_pct, y10_pct: strict numbers

every field required

the model loads once, checked

an input log: date, inputs, value

```
422 [(['body', 'y10_pct'], 'Field required')]
requirements_for: ['fastapi==0.143.0', 'pandas==3.0.6', 'pydantic==2.13.5']
```

requirements_for reads imports. The model file also needs scikit-learn and joblib.

The held-out days arrive as requests

939

requests, one per held-out day, in date order; every one status 200

4

distinct fitted values returned

16

leaves the tree could return

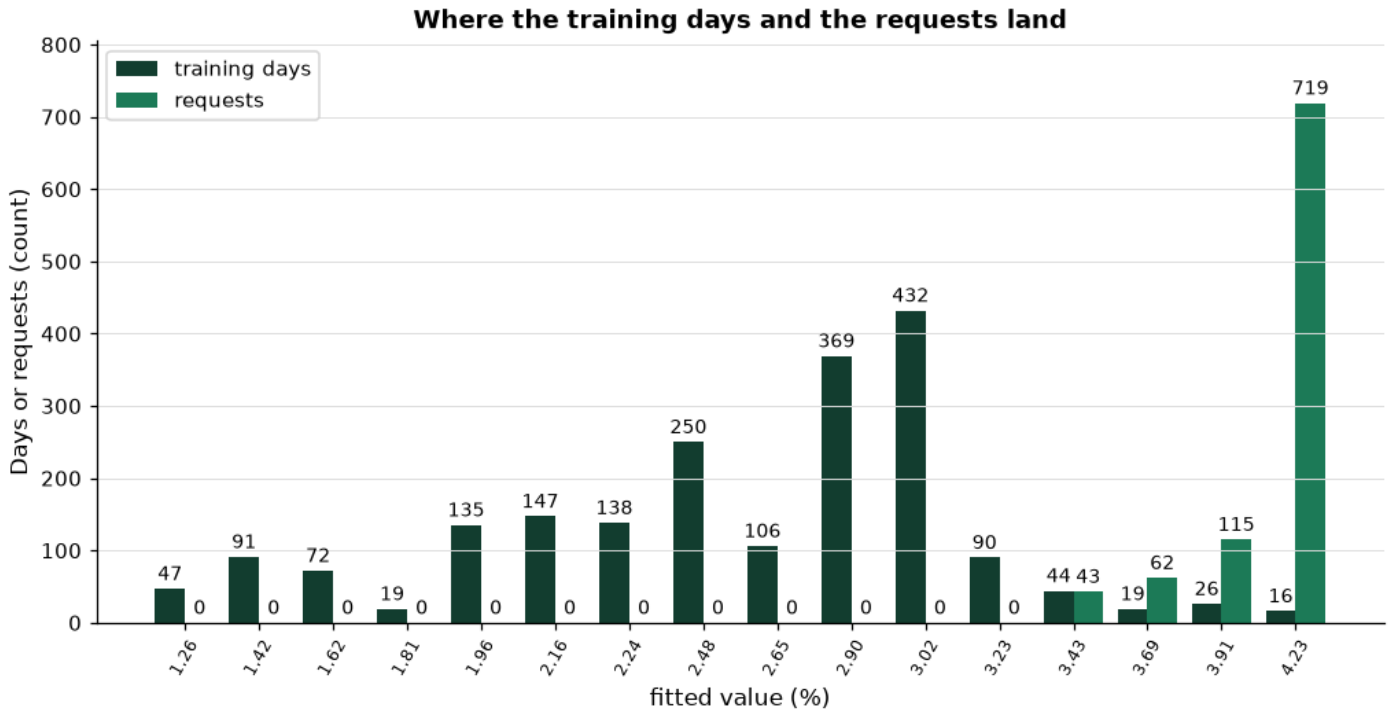
The input log

One JSON line a request: the date, the 2 Yr, the 10 Yr, the value returned.

Nothing about who sent it. Day 12's request log still holds no body.

A tree returns one of its leaves: on these days, only a few of them.

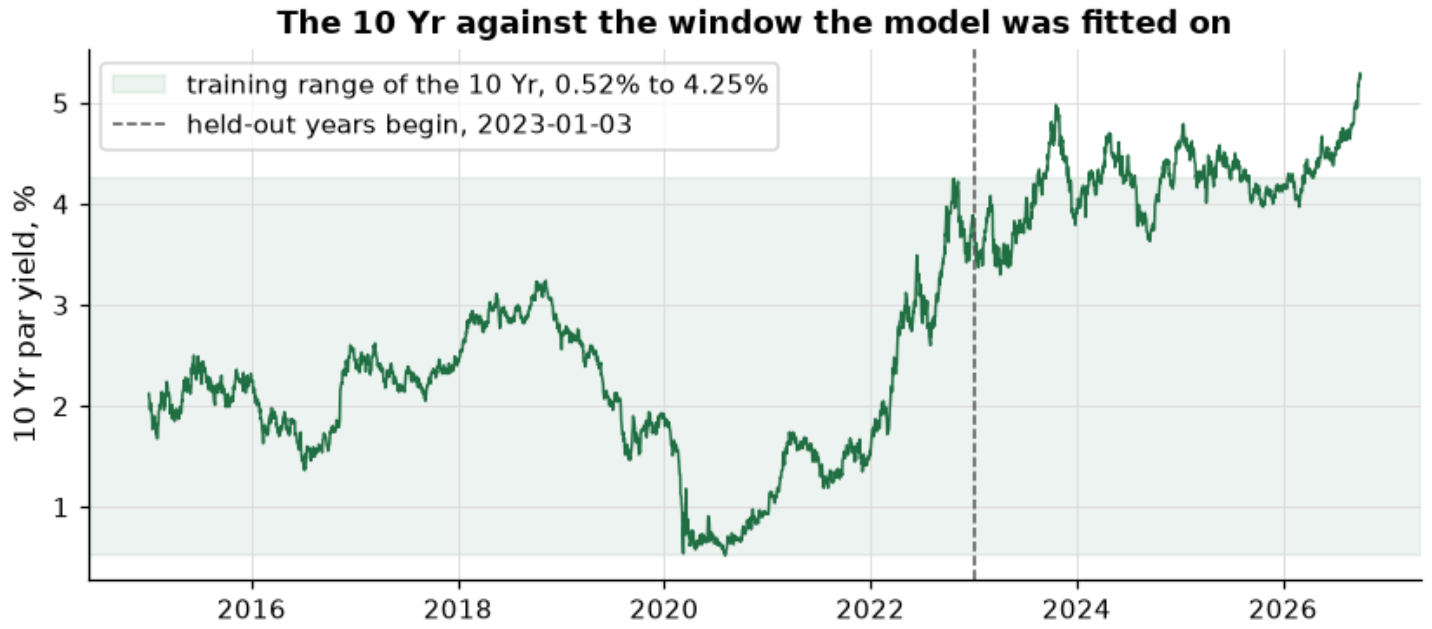
Where the days land



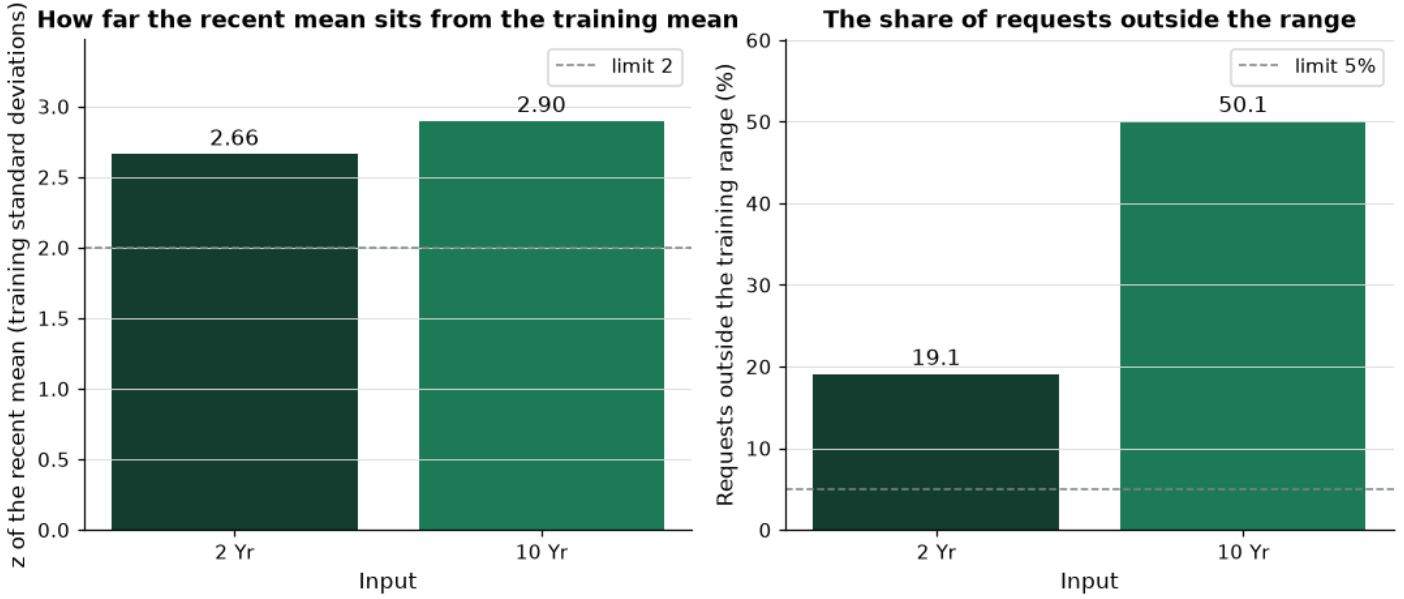
The days it was fitted on spread over all 16 values; the requests use only 4.

Drift: the inputs against the training window

	train_mean	train_std	train_min	train_max	recent_mean	z	outside_share
2 Yr	1.379	1.061	0.09	4.72	4.201	2.660	0.191
10 Yr	2.079	0.739	0.52	4.25	4.218	2.895	0.501



The drift against the limits



Input	z	Outside share
2 Yr	2.660	0.191
10 Yr	2.895	0.501

Both inputs sit over both default limits: a z above 2, and more than 5% outside the range.

Flags, and the error by quarter

Rule: flag when |z| > 2 or more than 5% of recent rows are outside the training range

29.4 bp

415 days with both inputs inside the training range

54.2 bp

524 days with an input outside it

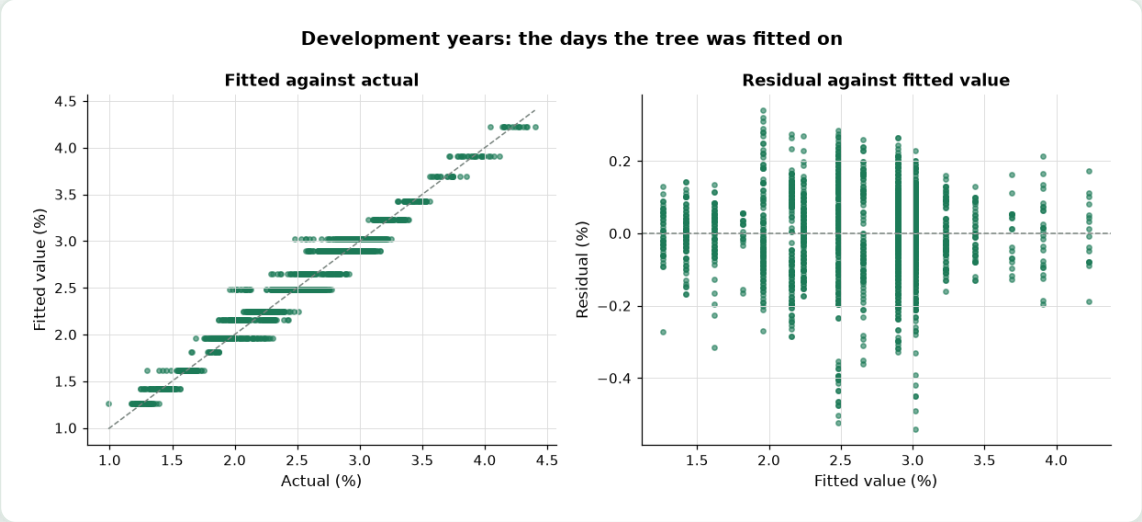
4.23%

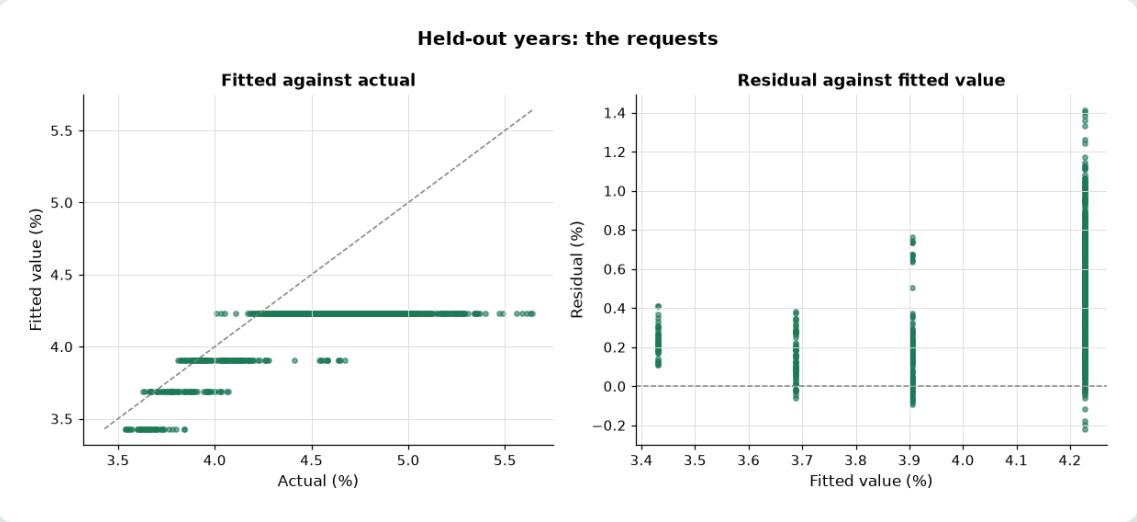
the highest value returned; the held-out 30 Yr reached 5.64%

```
quarter   days  outside_share  mae_bp
2023Q1    62           0.16    9.53
2023Q2    63           0.08   17.85
2025Q4    62           0.00   53.26
2026Q2    63           1.00   73.25
2026Q3    64           1.00   99.95
```

The error grows where the inputs leave the range. Inside the range is not a guarantee.

The tree on the requests

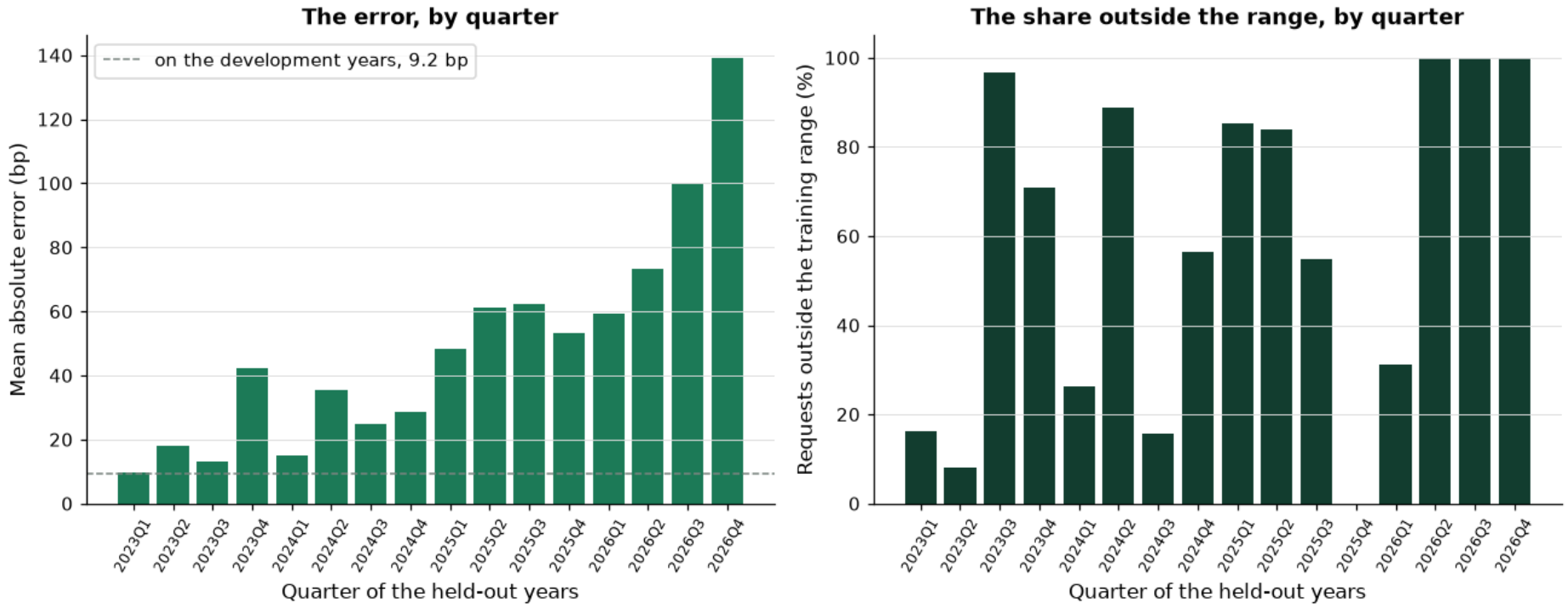




Rows	RMSE	MAE	MAPE (%)	R-squared
development years	0.1195	0.0921	3.7599	0.9598
held-out years	0.5163	0.4325	9.0776	-0.4292

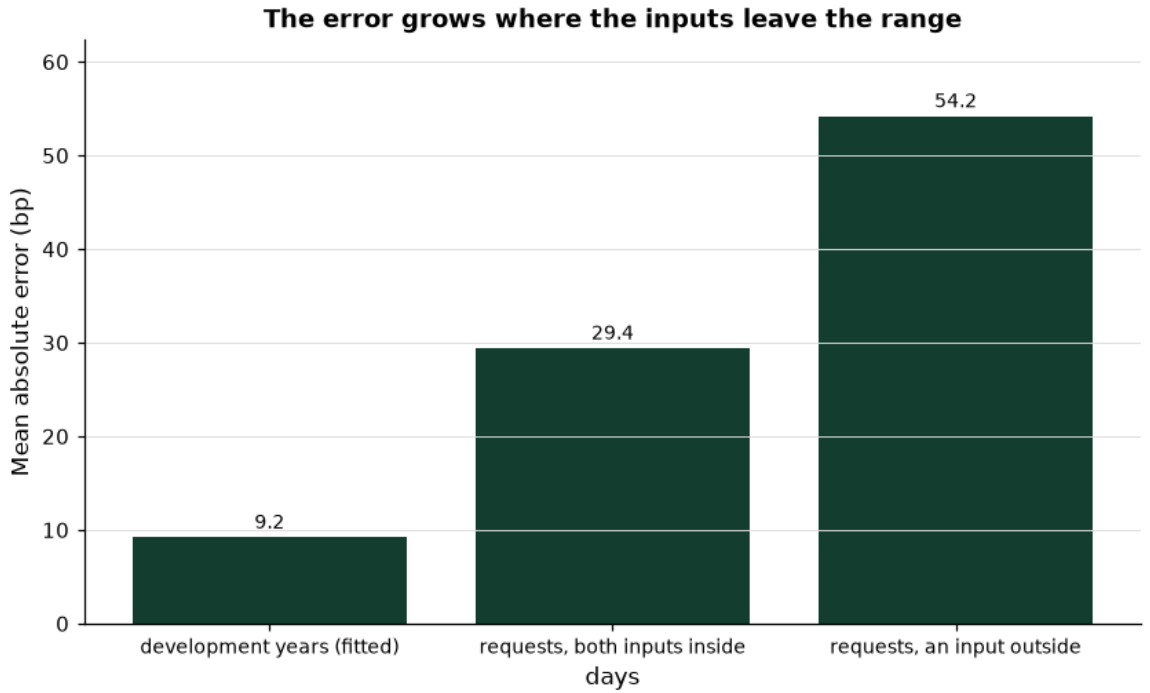
On the requests the fitted values stop at a ceiling, and R-squared is below zero. RMSE and MAE in percentage points.

The error by quarter



One quarter has nothing outside the range, and its error is still well above the first quarter's: inside the range is not a guarantee.

Inside and outside the range



Days	Count	Error (bp)
fitted on	2,001	9.2
inside range	415	29.4
outside range	524	54.2

The error grows where the inputs leave the training range.

What drift does not say

Not why

the inputs moved: a question about the economy

Not next quarter

no view on rates, no forecast

Not wrong today

a flagged input can land in a leaf that fits

Not a test passed

pytest checks code, not the world

What to do about drift is a decision for the people who use the service.

Excel and Git

Excel side by side (tested in Excel)

`=STANDARDIZE(AVERAGE(recent!C2:C940),AVERAGE(train!C2:C2002),STDEV.S(train!C2:C2002))`	2.895069
`=STANDARDIZE(..., STDEV.P(train!C2:C2002))`	2.895793
`=COUNTIFS(recent!C2:C940,">"&MAX(train!C2:C2002))+COUNTIFS(...,"<"&MIN(...))`	470

```
.gitignore:6:*.joblib   model/model.joblib
under model/ on the remote: ['model/model_card.json']
any .joblib file on the remote: no
```

The card is committed and pushed. The model file is ignored, with the reason beside the line.

The AI check

The docstring is the prompt. The deploy code that comes back runs and looks finished.

1

Docstring

Units, conventions, and what the code must never let through: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Plant the case the docstring promises to catch. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the reference pools the training rows and the requests: the 10 Yr's z falls to 1.23, and no flag.

Code that runs is not the same as code that is right.

What you can do now

- Save a model file with a card; load it only if its SHA-256 matches
- Never load a joblib or pickle file someone else made
- Serve the fitted value with its window, and log its inputs
- Measure drift against the training rows only
- Commit the card, never the file

Week 3, Day 4 Secrets, limits, and a cost meter